LUXEHAVEN

FEATURE #2 BUILD DOCUMENT

Product Catalogue + Search + Filtering

Field	Detail
PRD ID	PRD-002
Feature Name	Product Catalogue + Search + Filtering
Builds On	PRD-001 — Homepage Foundation (must be complete)
Priority	P0 — MUST HAVE
Sprint	Sprint 2 — Week 2
Enables	PRD-003 (Auth), PRD-004 (Checkout), PRD-009 (StyleMate AI)
Markets	United States (USD) + United Kingdom (GBP)

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PART A — PRODUCT REQUIREMENTS DOCUMENT (PRD-002)

1. Business Objective

Build the complete product discovery experience — the core of the LuxeHaven store. Customers must be able to browse all products, filter by category/price/rating/availability, search instantly, view full product details, and quick-view without leaving the catalogue. The admin must be able to add, edit, and manage all products from the backend dashboard.

2. User Stories

ID	As a...	I want to...	So that...
US-001	Customer	Browse all products in a clean grid	I can discover items I want to buy
US-002	Customer	Filter by category, price, rating, stock	I find exactly what I need quickly
US-003	Customer	Sort products by price, newest, rating	I can prioritise what matters to me
US-004	Customer	Search by typing a keyword	I find specific products instantly
US-005	Customer	See instant search results as I type	I don't have to wait or press Enter
US-006	Customer	Click a product to see full details	I can decide whether to buy it
US-007	Customer	Quick-view a product without leaving page	I can check details without losing my place
US-008	Customer (UK)	See prices in GBP with VAT note	I know exactly what I'll pay
US-009	Customer (US)	See prices in USD with correct state tax note	I understand the total cost
US-010	Admin	Add and edit products with all details	I can manage the full catalogue
US-011	Admin	Upload multiple product images	Products look professional and attractive
US-012	Admin	Set stock quantities and variants	Inventory is tracked accurately

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3. Functional Requirements

3.1 Pages & Components

Component / Page	Route	Description
Product Catalogue	/products	Full grid with filters, sort, pagination
Category Page	/products/category/[slug]	Pre-filtered catalogue per category
Product Detail	/products/[slug]	Full product page with gallery, variants, reviews
Search Results	/search?q=...	Search result grid with count
Search Bar	Global (header)	Instant search dropdown — 300ms debounced
Filter Sidebar	/products (sidebar/drawer)	All filters — desktop sidebar, mobile drawer
Quick View Modal	Overlay on catalogue	Mini product view without navigating away
Product Card	Reusable component	Used on catalogue, homepage, search, related
Admin Products List	/admin/products	Table with all products, search, bulk actions
Admin Product Form	/admin/products/new + /edit	Full add/edit form with image upload

3.2 Filter Requirements

Filter	Type	Details	URL Param
Category	Multi-checkbox	Jewellery, Clothing, Purses & Bags, Beauty	?category=jewellery
Price Range	Dual range slider	Min/max with manual input. USD or GBP.	?minPrice=10&maxPrice=200
Star Rating	Button select	4★+, 3★+, 2★+ options	?rating=4
Availability	Toggle switch	Show in-stock products only	?inStock=true
Badges/Tags	Multi-checkbox	New, Sale, Trending, Free Shipping	?tags=new,sale
Sort	Dropdown	Featured, Newest, Price asc/desc, Rating, Popular	?sort=price-asc
Page	Pagination	24 per page, numbered + prev/next	?page=2

3.3 Product Detail Requirements

- Image gallery: main image + up to 6 thumbnails, click to switch, hover zoom on desktop
- Dual pricing: USD and GBP stored separately, displayed based on locale cookie

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- Variant selection: size buttons (out-of-stock greyed), colour swatches
- Stock status: In Stock / Low Stock (< 5) / Out of Stock — shown prominently
- Quantity selector: 1–10, disabled if out of stock
- Delivery estimate: '7–12 business days' + 'No UK Customs Fees' for GBP users
- Accordion sections: Description, Shipping & Returns, Size Guide, Materials & Care
- Related products: 4 products from same category, using ProductCard component
- SEO: JSON-LD Product schema, dynamic meta title/description, canonical URL

3.4 Admin Product Management

- Products table: sortable columns, search, status filter, pagination
- Bulk actions: Activate / Deactivate / Delete (with confirmation modal)
- Add/Edit form: all product fields including dual pricing, variants, images, SEO
- Image upload: drag-and-drop, up to 8 images, reorderable, Cloudinary integration
- Slug: auto-generated from name, manually editable, uniqueness validated
- Seed data: 12 realistic sample products pre-loaded across all categories

4. Non-Functional Requirements

Requirement	Target
Product list load time	< 1.5 seconds (server-rendered with skeleton fallback)
Instant search response	< 200ms from API, debounced 300ms client-side
Image optimisation	Next.js Image with WebP, correct sizes prop, lazy loading
Mobile layout	Perfect at 375px (1 col), 640px (2 col), 1024px (3 col), 1440px (4 col)
Filter URL persistence	All filters reflected in URL — browser back works correctly
SEO — Product pages	Lighthouse SEO 95+, JSON-LD schema, dynamic meta tags
Accessibility	WCAG 2.1 AA — keyboard navigable filters, ARIA labels on all controls
TypeScript	Strict mode — all product types defined in types/product.ts

5. Database Schema Summary

Model	Key Fields	Relations
Product	id, tenantId, slug, name, retailPriceUSD, retailPriceGBP, stockQty, status, images, tags	belongs to Category, has many Variants, Reviews

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ProductVariant	id, productId, name, value, stockQty, priceAdjUSD, priceAdjGBP	belongs to Product
Category	id, tenantId, slug, name, description, image, sortOrder	has many Products
Review	id, productId, rating, body, videoUrl, images, verified	belongs to Product

6. API Endpoints Summary

Method	Endpoint	Auth	Description
GET	/api/products	Public	List with full filter, sort, paginate, facets
GET	/api/products/:slug	Public	Single product with variants + related
GET	/api/products/search?q=	Public	Lightweight instant search (5 results)
GET	/api/categories	Public	All categories with product counts
POST	/api/admin/products	Admin JWT	Create new product
PUT	/api/admin/products/:id	Admin JWT	Update product
DELETE	/api/admin/products/:id	Admin JWT	Soft delete product
POST	/api/admin/products/:id/images	Admin JWT	Upload image to Cloudinary

7. Acceptance Criteria

ID	Criteria	Pass Condition
AC-001	Product grid displays 24 products per page	Count verified on /products
AC-002	Filter by category works correctly	Only jewellery products shown when jewellery selected
AC-003	Price range slider filters correctly	Products outside range not shown
AC-004	Search returns results within 300ms	Stopwatch test on instant search
AC-005	URL updates on every filter change	Browser URL shows ?category=jewellery etc
AC-006	Product detail page loads all sections	Gallery, variants, accordion all present
AC-007	Quick View modal opens and closes correctly	Opens on eye icon, closes on Escape/backdrop

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AC-008	Admin can create a product and it appears in catalogue	End-to-end create → view test
AC-009	UK user sees GBP prices on all product pages	VPN UK test — all prices in £
AC-010	12 seed products visible in catalogue	Count on /products page
AC-011	Loading skeleton shows while fetching	Throttle network to Slow 3G — skeletons visible
AC-012	Product JSON-LD schema present on detail pages	Google Rich Results Test passes

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PART B — BOLT.NEW PROMPT (READY TO PASTE)

8. Bolt.new Build Prompt — Feature #2

Copy everything inside the box below and paste it into Bolt.new as your next prompt. Make sure Feature #1 is already built in your Bolt project before running this.

```
BOLT.NEW PROMPT — FEATURE #2: PRODUCT CATALOGUE + SEARCH + FILTERING
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You are a senior full-stack architect. Write production-ready code only —
no shortcuts, no placeholder logic, no TODO comments. After generating all
code, self-review for TypeScript errors, missing error handling, hardcoded
secrets, and mobile responsiveness. Fix anything before presenting output.

Feature #1 (Homepage Foundation) is already built. You are now adding
Feature #2: Product Catalogue, Search, and Filtering as a new module.
Do NOT modify or break any existing Feature #1 code. Only add new files
and extend existing ones where absolutely necessary (e.g. nav links).

=== WHAT TO BUILD — FRONTEND ===

1. PRODUCT CATALOGUE PAGE (/products)
   - Full product listing page with responsive grid layout
   - Grid: 4 columns desktop, 3 tablet, 2 mobile, 1 on small mobile
   - URL supports query params: /products?category=jewellery&sort=price-asc&page=2
   - Breadcrumb navigation: Home > Products > [Category]
   - Results count shown: 'Showing 1-24 of 248 products'
   - Sort dropdown: Featured, Newest, Price: Low to High, Price: High to Low,
     Best Rated, Most Popular
   - View toggle: Grid view / List view (persisted in localStorage)
   - Pagination: numbered pages + Previous/Next. 24 products per page.
   - Empty state: friendly message + 'Browse All Products' CTA when no results

2. PRODUCT CARD COMPONENT (components/shop/ProductCard.tsx)
   Required fields on each card:
   - Product image (Cloudinary optimised via Next.js Image)
   - Product name (truncated at 2 lines with ellipsis)
   - Price in correct currency (USD/GBP based on locale cookie)
   - Original price struck through if on sale + 'SALE' badge
```

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```
   - Star rating (1-5) with review count: e.g. ★★★★☆ (128)
   - Category badge (top-left corner)
   - Status badges: NEW (green), SALE (red), TRENDING (orange), OUT OF STOCK (grey)
   - Quick actions on hover: Wishlist heart icon, Quick View eye icon
   - 'Add to Cart' button (disabled + greyed if out of stock)
   - Smooth hover animation: card lifts with subtle shadow
   - Clicking card navigates to /products/[slug]

3. PRODUCT DETAIL PAGE (/products/[slug])
   - Image gallery: main image + thumbnail strip (up to 6 images)
     Click thumbnail to switch main image. Zoom on hover (desktop).
   - Product title (h1), category breadcrumb, SKU
   - Price display: current price large, original struck through if sale
     UK users: price shown with VAT included label
   - Star rating summary + total reviews count (links to reviews section)
   - Product description (rich text rendered, expandable if > 300 chars)
   - Variant selectors:
     * Size selector (if applicable): button grid, out-of-stock sizes greyed
     * Colour selector (if applicable): colour swatches, selected has ring
   - Quantity selector: minus/plus buttons, min 1, max 10
   - Add to Cart button (primary, full width on mobile)
   - Add to Wishlist button (secondary)
   - Trust signals below button: Free shipping threshold, returns policy,
     secure checkout badge, estimated delivery: '7-12 business days'
   - Product details accordion: Description, Shipping & Returns, Size Guide,
     Materials & Care
   - 'You May Also Like' section: 4 related products from same category
   - Customer Reviews section (placeholder with sample data for now)
   - Structured data: JSON-LD Product schema with name, price, image,
     description, sku, availability, aggregateRating

4. CATEGORY PAGES (/products/category/[category])
   Same layout as /products but pre-filtered to that category.
   Category header: category name, description, product count, hero image.
   Categories: jewellery, clothing, purses-bags, beauty, sale, new-arrivals

5. SEARCH RESULTS PAGE (/search?q=...)
   - Search results grid (same ProductCard component)
   - 'X results for [query]' heading
   - If 0 results: 'No results for [query]' + spelling suggestions +
     'Try these popular categories' links
   - Search term highlighted in product names in results

6. SEARCH BAR COMPONENT (components/shop/SearchBar.tsx)
   - Expands on click in header (already in Feature #1 header)
   - Instant search dropdown: shows top 5 matching products as user types
     (debounced 300ms). Each result shows: thumbnail, name, price.
   - 'See all X results for [query]' at bottom of dropdown
```

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```
   - Recent searches stored in localStorage (last 5)
   - Popular searches shown when bar is focused but empty
   - Press Enter or click result to navigate
   - Keyboard navigation: arrow keys move through results, Escape closes

7. FILTER SIDEBAR (components/shop/FilterSidebar.tsx)
   - Desktop: sticky left sidebar (280px wide)
   - Mobile: slide-in drawer from left, triggered by 'Filter' button
   - 'Clear All Filters' button at top when any filter is active
   - Active filter count badge on mobile 'Filter' button
   Filter sections (each collapsible with smooth animation):
   a) CATEGORY: Checkbox list — Jewellery, Clothing, Purses & Bags, Beauty
      Show product count per category in brackets
   b) PRICE RANGE: Dual-handle range slider (min/max)
      Manual input boxes for exact values
      US: $0 — $500+ | UK: £0 — £400+
   c) RATING: Star filter buttons — 4★ & up, 3★ & up, 2★ & up
   d) AVAILABILITY: Toggle — Show In Stock Only
   e) TAGS/BADGES: Checkbox — New Arrivals, On Sale, Trending, Free Shipping
   f) COLOUR: Colour swatch grid (if products have colour variants)
   - All filters update URL query params and results instantly (no page reload)
   - Filters persist when navigating back (from URL params)

8. QUICK VIEW MODAL (components/shop/QuickViewModal.tsx)
   - Opens when user clicks eye icon on product card
   - Modal with product image (left), details (right)
   - Simplified version of product detail: image, name, price, variants,
     quantity, Add to Cart, View Full Details link
   - Close on backdrop click or Escape key
   - Smooth open/close animation (Framer Motion)

=== WHAT TO BUILD — BACKEND ===

9. PRISMA SCHEMA — Add these new models to existing schema.prisma:

   model Product {
     id            String   @id @default(cuid())
     tenantId      String
     slug          String   @unique
     name          String
     description   String?  @db.Text
     shortDesc     String?
     sku           String?
     status        ProductStatus @default(DRAFT)
     categoryId    String
     supplierId    String?
     supplierSku   String?
     supplierPrice Decimal? @db.Decimal(10,2)
```

```
  costPrice     Decimal? @db.Decimal(10,2)
  retailPriceUSD Decimal @db.Decimal(10,2)
  retailPriceGBP Decimal @db.Decimal(10,2)
  salePriceUSD  Decimal? @db.Decimal(10,2)
  salePriceGBP  Decimal? @db.Decimal(10,2)
  stockQty      Int      @default(0)
  trackInventory Boolean @default(true)
  weight        Float?
  images        Json     @default('[]')
  tags          String[]
  isFeatured    Boolean  @default(false)
  isTrending    Boolean  @default(false)
  isNewArrival  Boolean  @default(false)
  metaTitle     String?
  metaDesc      String?
  avgRating     Float    @default(0)
  reviewCount   Int      @default(0)
  soldCount     Int      @default(0)
  createdAt     DateTime @default(now())
  updatedAt     DateTime @updatedAt
  category      Category @relation(fields: [categoryId], references: [id])
  variants      ProductVariant[]
  reviews       Review[]
}

model ProductVariant {
  id          String  @id @default(cuid())
  productId   String
  name        String
  value       String
  priceAdjUSD Decimal @default(0) @db.Decimal(10,2)
  priceAdjGBP Decimal @default(0) @db.Decimal(10,2)
  stockQty    Int     @default(0)
  sku         String?
  product     Product @relation(fields: [productId], references: [id], onDelete: Cascade)
}

model Category {
  id          String    @id @default(cuid())
  tenantId    String
  name        String
  slug        String    @unique
  description String?
  image       String?
  parentId    String?
  sortOrder   Int       @default(0)
  isActive    Boolean   @default(true)
  products    Product[]
```

```
    }

    model Review {
      id         String   @id @default(cuid())
      tenantId   String
      productId  String
      userId     String?
      authorName String
      rating     Int
      title      String?
      body       String   @db.Text
      videoUrl   String?
      images     String[]
      verified   Boolean  @default(false)
      helpful    Int      @default(0)
      createdAt  DateTime @default(now())
      product    Product  @relation(fields: [productId], references: [id], onDelete: Cascade)
    }

    enum ProductStatus { DRAFT ACTIVE ARCHIVED OUT_OF_STOCK }

  10. NESTJS — Products Module (src/modules/products/)
      Create full CRUD module:

      GET  /api/products
           Query params: category, search, minPrice, maxPrice, rating,
           inStock, tags, sort, page, limit, currency
           Returns: { products, total, page, totalPages, facets }
           facets = { categories with counts, priceRange: {min,max} }

      GET  /api/products/:slug
           Returns full product with variants, category, related products (4)

      GET  /api/products/search?q=&limit=5
           Instant search — returns top 5 matches with id, name, slug,
           image, priceUSD, priceGBP only (lightweight for dropdown)

      GET  /api/categories
           Returns all active categories with product counts

      POST /api/admin/products         (Admin JWT required)
      PUT  /api/admin/products/:id     (Admin JWT required)
      DELETE /api/admin/products/:id   (Admin JWT required)
      POST /api/admin/products/:id/images  (Cloudinary upload)

      All list queries must support:
      - Full text search on name, description, tags
```

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```
    - Price range filter (USD or GBP based on currency param)
    - Category filter (by slug)
    - Rating filter (gte value)
    - In-stock filter
    - Tag filter
    - Sort: newest, price-asc, price-desc, rating, popular
    - Pagination with correct total counts

11. ADMIN PANEL — Products Management (/admin/products)
    - Products list table: image thumbnail, name, category, price USD/GBP,
      stock qty, status badge, rating, actions (Edit/Delete/View)
    - Search and filter in admin table
    - Bulk actions: Activate, Deactivate, Delete selected
    - 'Add New Product' button → /admin/products/new
    - Add/Edit Product Form (/admin/products/new and /admin/products/[id]/edit):
      * Basic info: name, slug (auto-generated from name, editable), category,
        short description, full description (rich text textarea)
      * Pricing: USD price, GBP price, sale prices (optional), cost price,
        supplier price
      * Inventory: stock qty, SKU, track inventory toggle, supplier SKU
      * Images: drag-and-drop multi-image upload (up to 8). Preview grid.
        First image = main image. Drag to reorder.
      * Variants: dynamic add/remove rows for Size variants and Colour variants
        Each variant has: name, value, stock qty, price adjustment USD/GBP
      * SEO: meta title, meta description, URL preview
      * Flags: Featured, Trending, New Arrival toggles
      * Status: Draft / Active / Archived
    - Seed 12 realistic sample products across all 4 categories
      (jewellery x4, clothing x3, purses x3, beauty x2)
      with realistic names, descriptions, prices, and placeholder Unsplash images

=== SEO REQUIREMENTS ===

- /products page: title 'Shop All Products | LuxeHaven', description with
  keywords: jewellery, clothing, purses, beauty, US, UK
- /products/[slug]: Dynamic generateMetadata() using product name, description.
  JSON-LD Product schema: name, image, description, sku, offers (price, currency,
  availability), aggregateRating
- /products/category/[category]: Dynamic title and description per category
- Canonical URLs on all product and category pages
- /search page: noindex meta tag (search pages should not be indexed)

=== PERFORMANCE REQUIREMENTS ===

- Product list page: use React Server Components where possible
- Images: Next.js Image with sizes prop for responsive loading
  sizes='(max-width: 640px) 50vw, (max-width: 1024px) 33vw, 25vw'
- Search debounced at 300ms — never fires on every keystroke
```

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```
- Filter changes update URL and re-fetch without full page reload (useRouter)
- Implement loading skeleton UI for product grid while data loads
  (animated grey placeholder cards — same dimensions as real cards)

=== INTEGRATION WITH FEATURE #1 ===

- Update existing header nav: category links now point to correct
  /products/category/[slug] pages
- Update existing homepage 'Trending Now' and 'New Arrivals' sections
  to pull from real API (GET /api/products?tags=trending&limit=8)
  and (GET /api/products?tags=new-arrival&limit=8)
- SearchBar component: connect to GET /api/products/search?q=
- Homepage category cards: link to correct category pages

=== QUALITY CHECKLIST ===

Before finishing, verify:
[ ] All TypeScript types defined — no 'any'
[ ] Filter sidebar works on mobile (drawer) and desktop (sidebar)
[ ] URL params update on every filter/sort change
[ ] Pagination works correctly with filters applied
[ ] Product detail page loads with correct data from API
[ ] Admin can add a product and it appears in the catalogue
[ ] 12 seed products are visible in the catalogue
[ ] Quick View modal opens, shows data, closes correctly
[ ] Search dropdown shows results after 300ms debounce
[ ] All images use Next.js Image component
[ ] Loading skeletons show while products are fetching
[ ] Product JSON-LD schema present on detail pages
[ ] Mobile layout correct at 375px, 768px, 1024px, 1440px
```

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PART C — KNOWLEDGE BASE UPDATE (ADD TO BOLT SETTINGS)

9. Knowledge Base Addition

After Feature #2 is built successfully, go to Bolt.new → Settings → Knowledge → scroll to the bottom of your existing Global Instructions → add the following lines at the end. This ensures all future features know the product data rules.

```
BOLT.NEW KNOWLEDGE BASE — ADDITION AFTER FEATURE #2
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## PRODUCT DATA RULES (add to existing Knowledge Base)

All product queries MUST filter by tenantId.
Products have dual pricing: retailPriceUSD and retailPriceGBP.
Always serve correct currency based on locale cookie.
Slugs must be unique, URL-safe, auto-generated from product name.
Images stored as JSON array of Cloudinary URLs in Product.images field.
First image in array = main display image.
StockQty <= 0 = Out of Stock — disable Add to Cart button.
avgRating and reviewCount updated via database trigger or service method
whenever a new Review is created or deleted.

## SEARCH RULES

Instant search endpoint (/api/products/search) must return in < 200ms.
Full catalogue search must support pagination and return facet counts.
Search is case-insensitive, searches name + description + tags fields.
Debounce all client-side search inputs at 300ms minimum.

## URL PATTERN RULES

Product pages: /products/[slug]
Category pages: /products/category/[category-slug]
Search: /search?q=[query]
All filters expressed as URL query params — never in session/state only.
This ensures shareable URLs and browser back button works correctly.
```

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SUMMARY & NEXT STEPS

Step	Action	Status
1	Paste Bolt Prompt (Part B) into your existing Bolt project	☐ To Do
2	Wait for Bolt to generate all files — review the output	☐ To Do
3	Test: browse /products, filter, search, product detail, admin	☐ To Do
4	Verify 12 seed products are visible in the catalogue	☐ To Do
5	Update Knowledge Base with Part C additions	☐ To Do
6	Come back for Feature #3: User Authentication + Accounts + Wishlist	☐ Next

💡 Pro Tip: When pasting into Bolt, if the prompt is too long and Bolt truncates it, split it into two pastes — first the Frontend section, then the Backend section. Tell Bolt 'Continue building Feature #2 — now add the backend' for the second paste.

🚀 After Feature #2 is complete you will have: a fully working product catalogue that customers can browse, filter, search and view — and an admin panel to manage all products. Feature #3 adds user accounts, login, and wishlists so customers can save and track their favourite items.

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